

Coinnect

Self-service cash conversion
and wallet access

UMasenso Hub Cohort 5
SmartTech • UM Tagum

Partial physical prototype



The campus payment problem

- Cash in the wrong form** A large bill may not suit a small payment.
- Cash and wallet mismatch** A user may hold cash when wallet funds are needed, or the reverse.
- An extra service stop** Finding change or wallet access can add time and effort.

Problem hypotheses. Frequency, inconvenience, and current alternatives require interviews.

Initial users and proposed host

Participant	Role in the campus pilot
Students and staff	Initial users and transaction-fee customers
Campus administration	Proposed host and deployment approver
Nearby merchants	Secondary beneficiaries and interview participants

Early adopters: users with recurring needs who accept a disclosed fee. No host commitment is claimed.

The Coinnect solution

First pilot priority

Philippine peso bill and coin conversion

Bill to coin • Coin to bill • Bill to bill

Designed for local operation
without internet connectivity

Wallet services

Cash-in and cash-out after
integration testing and
operating readiness

Later capability

Foreign exchange

A guided transaction

01 Choose and review

Service, available amount,
output denominations, and fee

02 Insert cash or pay

Count accepted funds or
verify the wallet payment

03 Complete the service

Release the available cash
or confirm the wallet transfer

04 Keep a reference

Receipt or claim reference
when funds remain owed

Intended experience. End-to-end physical reliability still requires validation.

Prototype progress and evidence gaps

Present in the project	Still to validate physically
Touchscreen flows, backend, and firmware	Repeatable transactions on the assembled kiosk
Inventory, transaction, and claim logic	Jams, partial payout, and recovery after power loss
Camera checks and tamper-control design	Measured recognition and security performance

Team-confirmed stage: partial physical prototype. No accuracy or field-reliability result is claimed.

Alternatives and proposed differentiation

Alternative to investigate	Coinnect hypothesis to test
Cashier or acquaintance for change	A dedicated place for denomination conversion
Available cash-in or cash-out outlet	A convenient campus location for wallet access
Separate services for each task	Combined access through one kiosk

Local availability, fees, and convenience need comparison. No exclusive technology advantage is claimed.

Transaction-fee business model

Completed transactions × service fee

Fee revenue

Less provider charges
and direct transaction costs
= contribution

Operating requirements

Maintenance and cash servicing
Site, power, connectivity, and support
Funded cash float and settlement needs

Pricing and demand are unvalidated. Customer funds and cash float are not fee revenue.

Proposed supervised campus pilot

01 Demand discovery

Interview students, staff, and merchants.
Review existing service alternatives.

02 Kiosk readiness

Verify cash conversion and failure recovery.
Agree hosting, cash custody, and support.

03 Supervised use

Start with cash conversion.
Review reliability, usage, fees, and cost.

One approved indoor, climate-controlled location. Wallet services require additional readiness.

Social benefit and how to measure it

Intended benefit	Pilot evidence
Less effort obtaining suitable cash	User-reported time and cost versus prior alternatives
Convenient cash and wallet access	Completed transactions and repeat use with consent
Clear outcomes when a service fails	Support requests, unresolved claims, and resolution time

Benefits are intended outcomes. Interviews and supervised use must establish whether they occur.

Development milestones

Milestone	Evidence for the next decision
User needs and service scope	Interview findings and acceptable-fee hypotheses
Integrated cash prototype	Recorded intake, payout, inventory, and interruption tests
Operating and pilot plan	Measured costs, host permission, and support process
Pilot review	Observed results support refinement or continued testing

Proposed sequence. Timing depends on testing, resources, and approvals.

Team and incubation request

Vince Ronald Maravillas

Team leader

Stunley Opeña

Raziel Renz Salamat

Members

BS in Computer Engineering
Department of Engineering Education
UM Tagum

v.maravillas.135438.tc@umindanao.edu.ph

Support requested

Customer and pricing validation

Cash handling and reliability
feedback

Pilot introductions and operating
guidance